

SOP-TK-140: Atmospheric Storage Tank Floor and Shell Inspection

Document owner: Mechanical Integrity Department **Effective date:** 2024-03-10 **Review cycle:** 36 months

Section 1. Purpose and Scope

Governs internal floor-scan and shell-thickness inspection of atmospheric storage tanks, and the API 653 fitness-for-service evaluation triggered by their results.

Section 2. Inspection Intervals

Clause 2.1 — Internal inspection interval is determined by the shorter of: (a) the API 653 calculated interval based on corrosion rate, or (b) 15 years, whichever is sooner.

Clause 2.2 — A tank with a documented leak, or an external indication of bottom failure, requires internal inspection within 90 days regardless of its scheduled interval.

Section 3. Floor Scanning

Clause 3.1 — Every internal inspection includes a full robotic MFL (magnetic flux leakage) floor scan, not a spot-check grid, unless the tank diameter is under 30 feet, in which case a manual UT grid per API 653 Section 6 is acceptable.

Section 4. Retirement Thickness Criteria

Clause 4.1 — Floor plate minimum remaining thickness (MRT) is evaluated against API 653 Section 4.4 criteria: 0.1 inch for the tank bottom, or the design-specified minimum, whichever governs.

Section 5. Engineering Review Threshold

Clause 5.1 — Floor plate minimum remaining thickness for atmospheric storage tanks must be evaluated against API 653 criteria following any internal floor-scan inspection.

Clause 5.4 — A robotic MFL floor scan result triggers mandatory engineering review whenever isolated pitting depth exceeds the general corrosion trend by more than 40%, even if the average MRT across the floor remains within acceptable limits — localized pitting is a distinct failure mode from general thinning and is not adequately represented by an averaged result.

Section 6. Repair and Return to Service

Clause 6.2 — A floor area found below MRT shall be repaired (patch plate or replacement) before the tank returns to service; a below-MRT finding shall not be deferred to the next inspection cycle under any circumstance.

Section 7. Records

Clause 7.1 — Inspection reports, MFL scan data, and any engineering review are filed to the Tank Integrity File and retained for the operating life of the tank plus 10 years.